
Day-Ahead Electricity Price Forecasting using a Spatio-Temporal Probabilistic Transformer (MEGATRON)

1 Introduction

The transition toward a decarbonized energy grid has fundamentally altered the dynamics of wholesale electricity markets. In the United Kingdom, the increasing penetration of intermittent renewable energy sources, primarily wind and solar, has introduced severe supply-side uncertainty. Electricity is a largely non-storable commodity that requires real-time balancing between supply and demand: any sudden change in meteorological conditions forces the grid to rely on expensive, carbon-emitting gas power plants (CCGTs). This dynamic creates a highly volatile market where prices can spike drastically or even drop below zero during periods of excess renewable generation. The motivation of this project is to develop a robust forecasting engine capable of anticipating price movements. Because traditional statistical and econometric models fail to capture these complex non-linear relations, modern approaches have shifted toward computational intelligence and deep learning (1; 2).

Wholesale electricity costs eventually trickle down to retail consumers and industrial manufacturers. High forecasting errors force grid operators to hold expensive reserve power, increasing the overall economic burden on the country. As noted in industry literature, even a 1% improvement in forecast accuracy can save large utilities millions of pounds annually (3). Accurate forecasts also allow companies to hedge against price spikes.

This project focuses on Day-Ahead Electricity Price Forecasting (EPF). The objective is to predict the wholesale price of electricity for every hour interval of the following day. We mainly use two distinct inputs: historic electricity prices, and exogenous variables such as historic weather conditions, oil, and gas prices. Expanding on (4), we introduce Megatron, a Spatio-Temporal Probabilistic Transformer (STPT) to model the complex non-linear relationships between localized meteorological data, global commodity prices.

2 Methodology

2.1 Analysis of the electricity market

Electricity prices are primarily determined by the balance between supply and demand, and by the cost of the marginal generation unit (i.e., the last power plant needed to meet demand, often gas-fired).

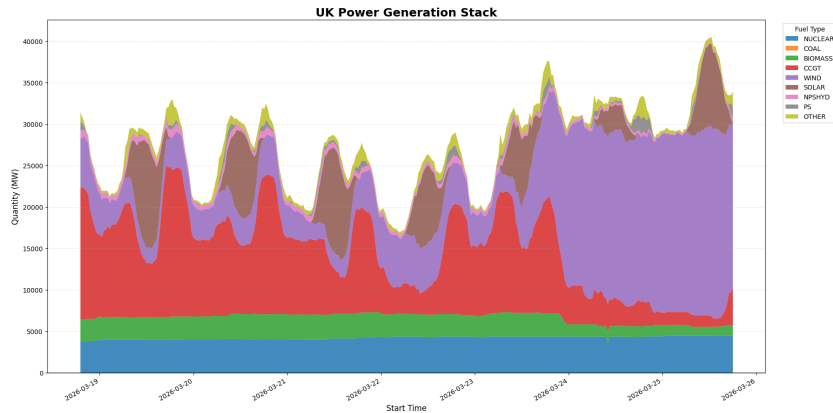


Figure 1: Data from Elexon

The electricity generation mix is composed of several technologies with distinct operational characteristics. Nuclear power and biomass constitute relatively stable and non-intermittent sources of generation. However,

they are inflexible, as their output cannot be rapidly adjusted in response to short-term fluctuations in demand. Similarly, electricity imports through interconnectors provide an additional source of supply, typically exhibiting moderate variability compared to domestic renewable generation.

On the demand side, electricity consumption exhibits both seasonal and intraday patterns. Annual seasonality is driven by heating demand in winter and cooling demand in summer. Additionally, there is strong daily and weekly seasonality: industrial and commercial activities dominate during working hours and weekdays, while residential demand typically peaks in the evening (see 2b). Electricity can see very sudden price spikes during system stress (e.g., a cloudy day with no wind during a winter evening cold wave, compounded by high gas prices).

In contrast, renewable energy sources such as wind and solar are inherently intermittent and constitute the main source of short-term variability on the supply side. Their output is directly driven by weather conditions, leading to significant fluctuations in available low-cost generation.

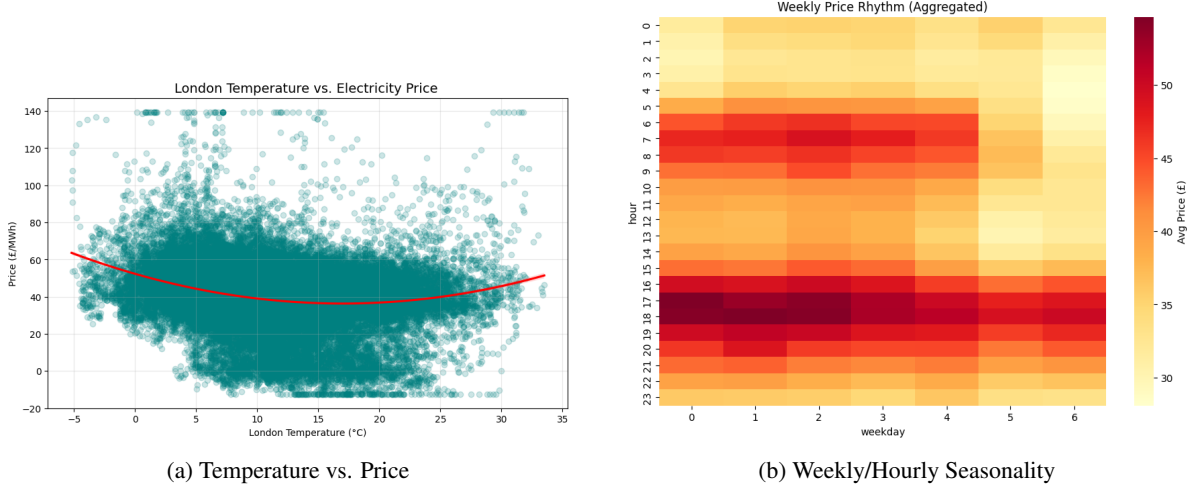


Figure 2: Analysis of electricity price drivers: (a) shows the “electricity smile” phenomenon during extreme temperatures, and (b) illustrates weekly and hourly seasonality trends.

Supply trends are mainly affected by wind and solar generation. As electricity is hard to store, gas power plants are used to complete the missing electricity demand due to their high flexibility. However, gas prices are volatile and can be largely affected by trade and geopolitical events (e.g., the war in Ukraine, or the closing of the Strait of Hormuz). This volatility is directly transmitted to electricity prices.

Electricity prices are characterized by high volatility and the occurrence of sudden price spikes. These spikes typically arise during periods of system stress, such as combinations of high demand and low renewable generation, particularly when coupled with elevated gas prices. Under such conditions, the system becomes increasingly reliant on expensive marginal generation, leading to sharp increases in prices.

2.2 Data collection and cleaning

We collected data from 2023 to 2026, including electricity prices using the Elexon API (10), spatial UK weather conditions (temperature at surface, wind speed at 100 m, cloud coverage) using the Open-Meteo API (11), and daily oil and gas prices using the yfinance library (12).

To construct a cohesive forecasting dataset, heterogeneous data sources with varying frequencies were aligned to a standard hourly resolution. Daily macroeconomic variables (oil and gas prices) were forward-filled, while 30-minute electricity prices and volumes were re-sampled to 1 hour intervals using the mean. Because Transformer architectures rely on strict, unbroken positional encodings, it is critical to maintain a continuous temporal sequence. Therefore, a small number of missing values in the electricity and volume series were resolved using linear interpolation. Electricity markets are inherently prone to extreme volatility. To manage anomalous price spikes and negative prices without erasing the vital market signals they represent, an asymmetric Interquartile Range (IQR) clipping strategy was applied. The lower bound was set to $3 \times \text{IQR}$ and the upper bound to $6 \times \text{IQR}$, preserving a wide enough distribution to train the model on heavy-tailed market events (and keep negative prices). To assist the model in capturing human behavioral patterns, temporal covariates (hour of the day, day of the week, and day of the year) were engineered.

The dataset was chronologically split into training (2023-03-02 to 2025-09-01), validation (2025-09-01 to 2025-12-31), and testing (2026-01-01 to 2026-03-01) sets. The test set was intentionally truncated at the beginning of March 2026 to isolate the model’s performance from the unpredictable exogenous macroeconomic shock caused by the closure of the Strait of Hormuz. Finally, to prevent vanishing or exploding gradients during backpropagation, all continuous features were normalized using a StandardScaler strictly fitted on the training set to prevent data leakage.

2.3 Baseline Models

Recent advancements in time-series forecasting have seen a transition from Recurrent Neural Networks (RNNs) like LSTMs toward Transformer-based architectures, which are better suited for integrating heterogeneous datasets (6).

We employ Linear Regression and XGBoost (5) as baselines (XGboost is often a very strong benchmark that is hard to beat especially for tabular data). For these baseline models, we used only a few specific weather points (London, Cornwall solar farm, North Sea wind farm).

2.4 Transformer Architecture

Megatron (STPT) is designed to capture complex dependencies in high-dimensional grid weather data and auxiliary scalar time series. The architecture consists of three primary stages: a deep residual spatial encoder, a temporal transformer backbone, and a probabilistic forecasting head.

Weather data is formulated as an 8x8 grid over the UK with 3 channels (wind, temperature, clouds). We use a 7-day lookback period (168 hours) and a 24-hour forecast horizon. The data is scaled using standard normalization techniques for faster training.

2.4.1 Spatial Feature Extraction (ResNet-CNN)

To process the CxHxW (3 x 8 x 8) meteorological grids, we employ a Deep Residual CNN Encoder. Unlike standard convolutional layers, our encoder utilizes Residual Blocks (7) to mitigate the vanishing gradient problem and allow for the preservation of fine-grained spatial features across layers. Each residual blocks uses 2 kernels of size 3x3, padding and batch normalization.

These blocks introduce identity shortcut connections that bypass parameterized layers, defined mathematically as:

$$y = \mathcal{F}(x, \{W_i\}) + x \quad (1)$$

where x and y are the input and output vectors, and $\mathcal{F}$ represents the residual mapping to be learned. This allows the model to capture complex, multi-scale localized weather phenomena (e.g, pressure gradients or localized storm fronts).

To ensure the model is robust to minor spatial shifts in weather patterns, the final spatial maps are aggregated using Adaptive Average Pooling. This yields a dense, fixed-size spatial feature vector $z_t \in \mathbb{R}^{64}$ for each hour t in the look-back window.

The convolutional layer is designed to extract localized spatial weather patterns that directly impact the energy grid. Specifically, it captures meteorological factors affecting both supply (such as wind speeds at offshore farms and solar irradiance at photovoltaic installations) and demand (such as cold temperatures and high winds in densely populated urban centers).

2.4.2 Heterogeneous Feature Fusion and Temporal Attention

At each time step t , the extracted spatial summary z_t is concatenated with the scalar variables $s_t \in \mathbb{R}^k$ (historical prices) to form a unified input vector $u_t = [z_t; s_t]$. This combined sequence is linearly projected into a high-dimensional latent space ($d_{model} = 84$).

Because self-attention mechanisms are inherently permutation-invariant, we inject standard sinusoidal positional encodings into the latent vectors to preserve the sequential temporal ordering of the data.

The core temporal processor is a standard Transformer Encoder block comprising 4 layers. Each layer utilizes Multi-Head Self-Attention (MHSA) with $H = 12$ heads (8). The self-attention mechanism allows the network to dynamically weigh the importance of historical time steps relative to one another, effectively capturing long-range dependencies:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (2)$$

where the Queries (Q), Keys (K), and Values (V) are linear projections of the fused historical grid and market states, and d_k is a scaling factor to prevent vanishing gradients. This enables Megatron to selectively "attend" to critical historical events (such as a severe weather anomaly days prior) that have a delayed impact on the forecasting horizon.

2.4.3 Dual-State Fusion and Probabilistic Inference

To generate the final forecast, we must compress the temporal sequence of hidden states into a single predictive representation. Rather than relying exclusively on the final hidden state h_T (which can introduce a recency bias), or utilizing a simple global average $\bar{h}$ (which flattens temporal momentum), we employ a Dual-State Fusion strategy. The model concatenates the final step h_T with the sequence mean $\bar{h}$, yielding a rich representation that encapsulates both immediate local momentum and global weekly context.

To account for the high volatility and non-Gaussian nature of energy and weather data, the fused representation is passed through a linear layer to output the parameters of a Student's t-distribution (9). The probabilistic head models three distinct parameters:

- **Degrees of Freedom (ν):** Controls the kurtosis (heavy-tailedness) of the distribution, making the model robust to outliers. Constrained via a softplus activation to ensure $\nu > 2$.
- **Location (μ):** The expected mean of the prediction.
- **Scale (σ):** The measure of variance, constrained to be strictly positive ($\sigma > 0$).

Megatron is trained end-to-end by minimizing the Negative Log-Likelihood (NLL) of the true target y_t under the predicted Student's t-distribution:

$$\mathcal{L}_{NLL} = -\log \left(\frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sigma\sqrt{\nu\pi}} \left(1 + \frac{(y_t - \mu)^2}{\nu\sigma^2} \right)^{-\frac{\nu+1}{2}} \right) \quad (3)$$

This objective function naturally penalizes the model for overconfidence during highly volatile periods, forcing it to learn a reliable estimate of its own uncertainty alongside the point forecast.

2.5 Model Training

Training was executed on an NVIDIA RTX 4060 GPU. We utilized the AdamW optimizer and gradient clipping optimizer for quicker convergence. We employed early stopping to mitigate overfitting alongside standardized dropout layers.

Table 1: Model Hyperparameters

Parameter	Value	Description
cnn_out	64	Output size from 8X8 CNN
d_model	84	Transformer hidden dimension
num_heads	12	Number of attention heads (8)
num_layers	4	Number of Transformer layers
dropout	0.3	Dropout rate

3 Results

To evaluate the forecasting performance, we utilized two standard regression metrics: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

MAE calculates the average absolute difference between the predicted and actual electricity prices. It provides a baseline accuracy whereas RMSE squares the errors before averaging them, which disproportionately penalizes large forecasting errors.

Table 2: Performance Metrics Comparison on Validation Set

Model	MAE	RMSE
Linear Regression	11.2187	14.6724
XGBoost	10.7547	14.5243
STPT	8.7711	12.5725

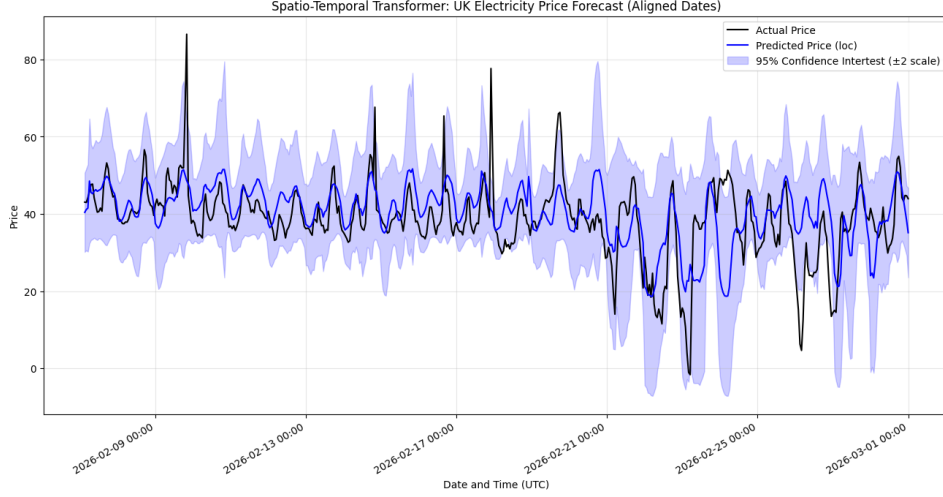


Figure 3: Model prediction time series on a sample of the test set.

To avoid selection bias, the different models and transformer hyper-parameters were compared on the validation set. The test set was used to evaluate the final architecture. On the test set, the model achieved a **Root Mean Square Error (RMSE) of 7.7** and a **Mean Absolute Error (MAE) of 5.5**. Machine learning models tend to gravitate towards the mean, hence our model struggles with extreme price spikes. Nevertheless these deviations are correctly captured by the increased uncertainty bounds generated by the Student-T head. The model achieved an interval coverage of 99.02% for the empirical 95% confidence interval on the test set implying that the model is not well calibrated. However, in a risk-averse trading environment, a highly reliable model is preferred, as the uncertainty signal tells the trader to stop trading and limit their risk. Future iterations of this model could implement a penalty term for interval width

4 Discussion

The dataset was restricted to the last three years of data to intentionally exclude the anomalous price volatility caused by the 2022 global energy crisis resulting from the war in Ukraine. While this provided a more stable baseline, a consequence of this limited timeframe is that the model lacks exposure to rare meteorological combinations, such as severe storm events.

Attempts to increase model complexity generally led to overfitting rather than performance gains. Various architectural modifications were tested, including increasing the attention dimensions, adding more attention heads and CNN layers, altering CNN kernel sizes, and extending the temporal lookback period. We also experimented with expanding the spatial grid to 11x11 and introducing additional weather variables (e.g., rain and wind direction). While a CNN can theoretically process much larger spatial grids to track complex moving weather systems (e.g., a storm front moving from the North Sea to London), doing so risks pivoting the network's focus toward redundant weather prediction rather than price forecasting. Finally, extended training regimes designed to observe a "double descent" phenomenon were also unsuccessful.

Predicting electricity prices remains a highly complex task. A significant methodological limitation of the current model is that it relies on actual historical weather data, rather than day-ahead weather forecasts, during the testing phase. In a real-world scenario, models must operate on imperfect forecasts: future work should either introduce synthetic noise into the next day weather data to better simulate this uncertainty or utilize true historical forecast archives. Furthermore, subsequent models could benefit from incorporating additional exogenous variables, such as generation availability (power plant outages), calendar effects (bank holidays), and cross-border power exchange dynamics (e.g., electricity prices and weather in interconnected markets like France, Ireland, and the Netherlands).

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